EXPERIMENT – 6

INDUSTRIAL HVAC CONTROL FIRMWARE: ADC TEMPERATURE ACQUISITION AND TIMER-INTERRUPT PWM FAN CONTROL

Code:

```
#define TEMP A0

#define FAN  8


volatile byte pwm = 0;

volatile byte count = 0;


ISR(TIMER1_COMPA_vect)

{

  count++;


  if (count >= 100)

    count = 0;


  digitalWrite(FAN, count < pwm);

}


void setup()

{

  Serial.begin(9600);

  pinMode(FAN, OUTPUT);


  TCCR1A = 0;
```

```
  TCCR1B = (1 << WGM12) | (1 << CS11);

  OCR1A = 199;

  TIMSK1 = (1 << OCIE1A);

  sei();
}

void loop()
{
  int adc = analogRead(TEMP);

  float temp = adc * 500.0 / 1023.0;

  if (temp < 25)
    pwm = 0;
  else if (temp < 30)
    pwm = 30;
  else if (temp < 35)
    pwm = 60;
  else
    pwm = 90;

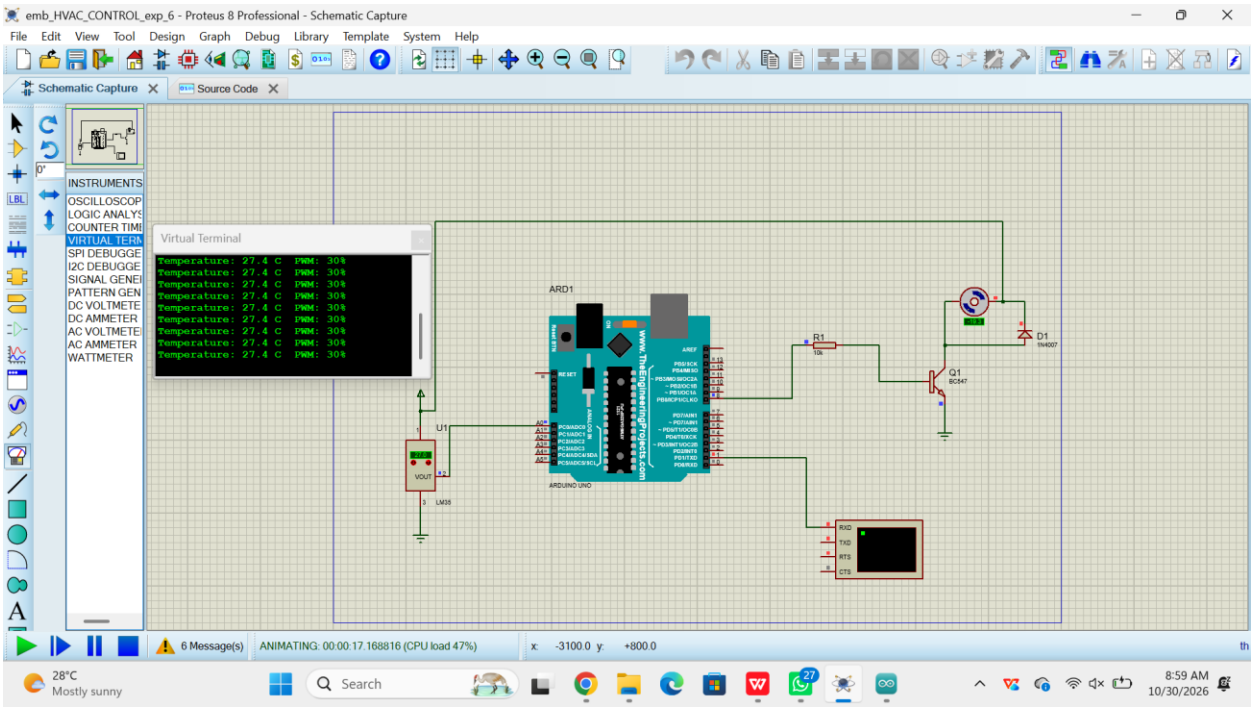
  Serial.print("Temperature: ");
  Serial.print(temp, 1);
```

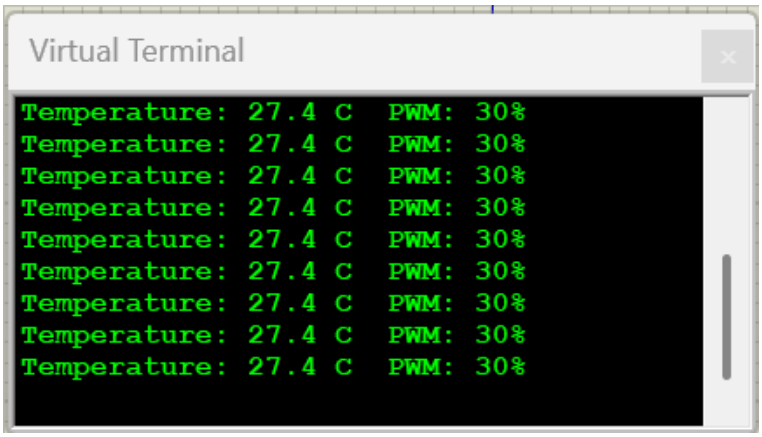
```
 Serial.print(" C  PWM: ");
 Serial.print(pwm);


 Serial.println("%");


 delay(1000);
}
```

Output:





Reflection: I learned about ADC-based temperature acquisition and PWM fan control. I understood timer interrupts and temperature sensor interfacing. I verified fan speed control based on temperature variations.